

5 A student uses a circuit containing an ammeter, a voltmeter and a cell to take measurements to determine the resistance of a length of nichrome wire.

(a) (i) Define resistance.

.....
..... [1]

(ii) Draw a circuit diagram to show how the components should be connected. Use the symbol for a resistor to represent the nichrome wire.

[2]

(b) The student also measures the length and the diameter of the wire. Table 5.1 shows the measurements recorded for each quantity.

Table 5.1

quantity	measurement
length	(0.864 ± 0.001)m
diameter	(0.496 ± 0.002)mm
voltmeter reading	(1.38 ± 0.02)V
ammeter reading	(0.276 ± 0.001)A

(i) Show that the resistance of the wire is 5.00 Ω.

[1]

(ii) Calculate, to three significant figures, the resistivity ρ of the nichrome.

ρ = Ω m [3]

(iii) Calculate the percentage uncertainty in ρ .

percentage uncertainty = % [2]

(iv) Determine the absolute uncertainty in ρ .

absolute uncertainty = Ω m [1]